The Fed Put in General Equilibrium

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Central bank puts, or the idea that central banks intervene to prop up asset prices in times of trouble, have been the object of a large and still growing literature in both modern macroeconomics and finance. As an opening salvo, Bernanke & Gertler (2001) explored a rich DSGE setting where the monetary authority reacts to asset prices directly, and showed how this policy is suboptimal. However, their approach leads to reactions to both shortfalls and windfalls in asset prices (closer to a Fed put-call spread); others in the macro literature have gone the same route. In finance, Cieslak & Vissing-Jorgensen (2021) document the Fed put, and recent work has explored its effects on asset prices.

1 Introduction

”Should the central banks react to asset prices?” has, in some way or another, been the opening line of a large and growing literature in both modern macroeconomics and finance. Bernanke & Gertler (2001) fired the opening salvo arguing against it, and from then on several others have followed and found benefits and costs to such a policy.

That the topic has attracted the attention of such luminaries is unsurprising. Monetary policy is one of the best tools in a policymakers kit, and its science one of the most important

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Início: policy, preocupação com bolhas, simulações (Bernanke & Gertler 1999, 2001, Gilchrist & Leahy 2002, Cechetti et al 2007, Bean 2004) desenvolvimentos: determinação (Bullard et al 2002, Calstrom & Fuerst 2007), OLG (Nistico 2012)

/bota aqui UwU:The Fed Put and Monetary Policy: An Imperfect Knowledge Approach (Ifrim 2021)

documentação Cieslak and Vissing-Jorgensen (2020)

actual fed PUT: Pastore 2025 finance tradition, concerned with equilibrium prices

This paper: closed form symmetric solution sheds light on previous results that were so far discussed but unproven asymmetric implementation and exploration of macro outcomes results: fed put beneficial at low reactivity levels. symmetric rules deliver the greatest gains, but assymetric rules deliver smaller losses if the reaction is too strong.

2 The Model

We develop a baseline New Keynesian model with a representative household with two modifications. First, we introduce a risky asset in the form of a Lucas tree. Second, we alter the monetary rule such that it may react to asset price shortfalls (the "Fed put"). We also solve a version of the model where the reaction is symmetric (echoing Bernanke (2001)), which would be better described as a "Fed put-call spread".

2.1 Households

A representative household maximizes

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t Z_t \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} \right), \tag{1}$$

where Z_t is a demand shock; subject to the budget constraint

$$C_t + M_t S_t + Q_t B_t \leq W_t N_t + B_{t-1} + (M_t + D_t) S_{t-1} + L_t + T_t, \tag{2}$$

where B_t and Q_t denote safe, government bond holdings and their prices, M_t the price of the Lucas tree, D_t its dividend, S_t its holdings, and L_t and T_t are lump sum profits and net fiscal transfers, respectively. Letting lowercase letters denote their real counterparts, first-order conditions yield:

$$Q_t = \beta \mathbb{E}_t \left[\frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}} \frac{1}{\Pi_{t+1}} \frac{Z_{t+1}}{Z_t} \right], \tag{3}$$

$$w_t = C_t^{\sigma} N_t^{\varphi}, \tag{4}$$

$$m_t = Q_t \mathbb{E}_t \left[\Pi_{t+1}(m_{t+1} + d_{t+1}) \right]. \tag{5}$$

Where $\Pi_{t+1} = \frac{P_{t+1}}{P_t}$ is the gross inflation for the $t+1$ period.

2.2 Firms

Final-good firms are competitive and aggregate differentiated goods using a Dixit-Stiglitz aggregator. Intermediate firms produce with decreasing returns:

$$Y_t(j) = A_t N_t(j)^{1-\alpha}, \tag{6}$$

and set prices subject to Calvo frictions. Standard arguments yield a New Keynesian Phillips Curve relating inflation to real marginal costs.

2.3 Monetary Policy

Let $\tilde{x}_t \equiv x_t - x_t^n$ denote the log difference of a variable to its natural level. The central bank follows a Taylor rule augmented with an asymmetric response to asset price shortfalls, subject to zero-mean gaussian shocks. Under zero inflation targeting, in logs:

$$i_t = r^{ss} + \phi_\pi \pi_t + \phi_y \tilde{y}_t + \phi_m \psi_t + u_t, \tag{7}$$

where

$$\psi_t = \min\{0, \tilde{m}_t\}. \tag{8}$$

This embeds the Fed put: monetary policy responds to asset prices only when they fall below equilibrium.

2.4 Exogenous Processes

Log productivity, real dividend yield, demand and monetary shocks follow processes:

$$a_t = \rho_a a_{t-1} + \varepsilon_t^a, \tag{9}$$

$$d_t = (1-\rho_d)d^{ss} + \rho_d d_{t-1} + \varepsilon_t^d, \tag{10}$$

$$z_t = \rho_z z_{t-1} + \varepsilon_t^z, \tag{11}$$

$$u_t = \rho_u u_{t-1} + \varepsilon_t^u \tag{12}$$

2.5 Equilibrium

Goods market clearing requires $Y_t = C_t$, and we normalize the tree's holding to one.

The natural rate is standard, given by

$$r_t^n = \rho + \sigma \mathbb{E}_t \Delta y_{t+1}^n - \mathbb{E}_t \Delta z_{t+1} \tag{13}$$

2.6 Log-linearized system

The equilibrium characterization that emerges is mostly standard:

IS curve

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\sigma}(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n) \tag{14}$$

Phillips curve

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t \tag{15}$$

Asset pricing

$$\tilde{m}_t = \mu \mathbb{E}_t[\tilde{m}_{t+1}] + (1-\mu)\mathbb{E}_t[\tilde{d}_{t+1}] - (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n) \tag{16}$$

Where $\mu = \frac{m^{ss}}{m^{ss}+d^{ss}}$

Policy rule

$$i_t = \rho + \phi_\pi \pi_t + \phi_y \tilde{y}_t + \phi_m \psi_t + u_t \tag{17}$$

$$\tag{18}$$

It is immediately clear that the tree's dividend and price do not impact consumption choices directly. The macro block is shielded from the financial block, but affects it via both interest rates and inflation. This is because the representative agent does not treat positive dividend shocks as wealth, anticipating future bad times (and vice versa). A future extension would be managing to create this interaction by means of a more sophisticate modeling of assets.

2.7 Occasionally binding structure

The presence of ψ_t introduces a nonlinearity. The model admits a piecewise-linear representation:

Regime 1 (no support):

$$\tilde{m}_t \geq 0 \Rightarrow \psi_t = 0$$

Policy reduces to standard rules.

Regime 2 (put active):

$$\tilde{m}_t < 0 \Rightarrow \psi_t = -\tilde{m}_t$$

Policy responds directly to asset price deviations.

Thus, equilibrium dynamics are governed by a system with occasionally binding constraints.

3 Solving the Symmetric Case

A simpler version of the monetary rule is achieved by allowing responses to both positive and negative asset price deviations from equilibrium. This has the added bonus of providing a benchmark and being more directly comparable to the previous literature. The Rational Expectations equilibrium becomes:

$$\text{IS}: \quad \tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\sigma}(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n) \tag{19}$$

$$\text{NKPC}: \quad \pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t \tag{20}$$

$$\text{AP}: \quad \tilde{m}_t = \mu \mathbb{E}_t[\tilde{m}_{t+1}] + (1-\mu)\mathbb{E}_t[\tilde{d}_{t+1}] - (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n) \tag{21}$$

$$\text{MR}: \quad i_t = \rho + \phi_\pi \pi_t + \phi_y \tilde{y}_t + \phi_m \tilde{m}_t + u_t \tag{22}$$

$$\tag{23}$$

Demand and productivity shocks enter only through the natural rate. Seeking a solution of the form

$$\tilde{y}_t = a_r r_t^n + a_u u_t + a_d d_t \tag{24}$$

$$\pi_t = b_r r_t^n + b_u u_t + b_d d_t \tag{25}$$

$$\tilde{m}_t = c_r r_t^n + c_u u_t + c_d d_t \tag{26}$$

We find that a nonzero ϕ_m appears with a positive sign on the denominator for all of the natural rate and monetary shocks coefficients, thus making them smaller; and it appears on both both numerator and denominator for a_d and b_d, making them positive (see Appendix A for a full proof).

As asset prices move due to both real (i.e natural rate) and financial (dividend) shocks, a monetary policy that responds to them has ambiguous effects on the economy. On the one hand, the reaction to real shocks is amplified, which reduces the variance of both $\tilde{y}_t$ and π_t (the denominator effect). On the other, financial shocks are now transmitted to the real economy when they would otherwise be neutral, which increases both variances.

These findings rationalize the simulations conducted by Bernanke & Gertler (2001). When discussing their results that reactions to asset prices may improve economic outcomes, they state:

> ”Our interpretation of this effect is as follows: A shock to stock prices may temporarily change the natural real rate of interest, a change that in principle should be accommodated by a fully optimal policy rule. Putting a small weight on stock prices therefore may help a bit, at least in some circumstances and on some dimensions”

By making an analytical solution more manageable, we prove their interpretation to be accurate and provide the precise mechanisms through which it manifests itself.

4 Policy Rules Comparison

We now move on to implementing the asymmetric, put-like reaction function. We do not provide a closed-form result, but rather simulate the model under standard calibrations for primitives (presented on Appendix B) and fixed coefficients for reactions to inflation and the output gap.

This allows for the numerical computation of both unconditional moments and impulse response functions. We then compare these to the closed-form results obtained for standard

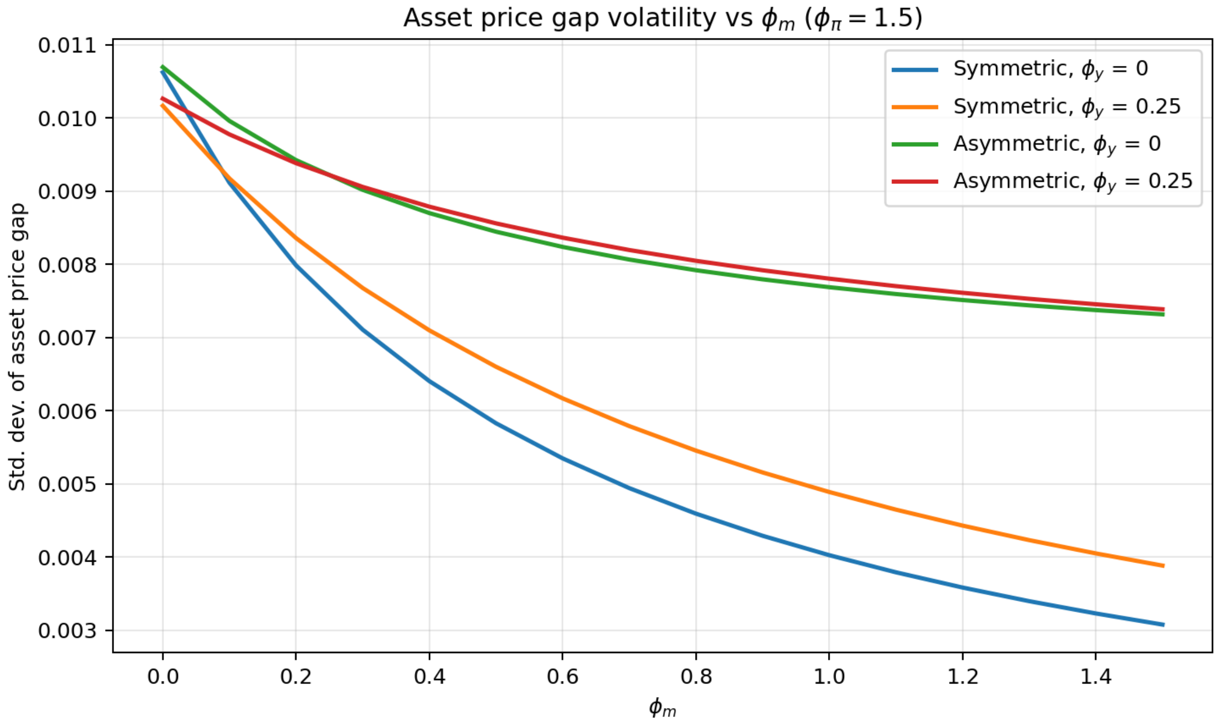


Figure 1: Asset price volatility different under monetary rules

and symmetric policy rules.

4.1 Standard Deviation Comparison

Figures 1 through 3 plot how the standard deviation for asset prices, inflation and the output gap varies as ϕ_m increases. For a better understanding of the effects of different policy rules, we implement four versions. The first, in blue in all figures, stands for a central bank that reacts only to inflation under $\phi_m = 0$ (which corresponds to the Y-axis), and then reacts symmetrically to asset prices under positive ϕ_m. For the second, in orange, we add set the reaction to output gap, ϕ_y, to 0.25 and retain the symmetric response to asset prices. The third, in green, and fourth, in red, are the asymmetric versions of the first and second respectively. As a first pass test of our implementation, we confirm that rules 1 and 3 and 2 and 4 deliver the same results under $\phi_m = 0$.

Asset price volatility is monotonically decreasing in ϕ_m, as we would expect. It is interesting that the *symmetric* rule delivers lower volatility. As it reacts to both positive and negative shocks, it compresses asset price movements more so than the asymmetric rule. This is not necessarily a good thing: even an investor that dislikes volatility would probably be happy if that volatility is heavily skewed upwards, after all.

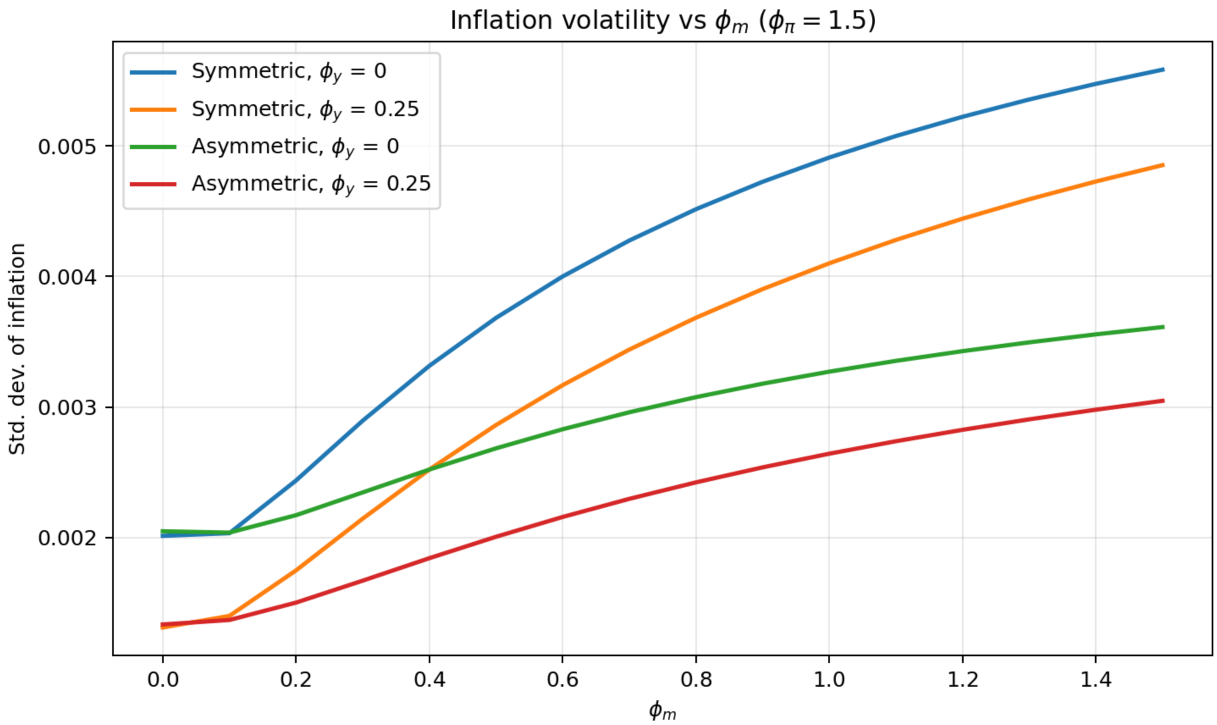


Figure 2: Inflation volatility different under monetary rules

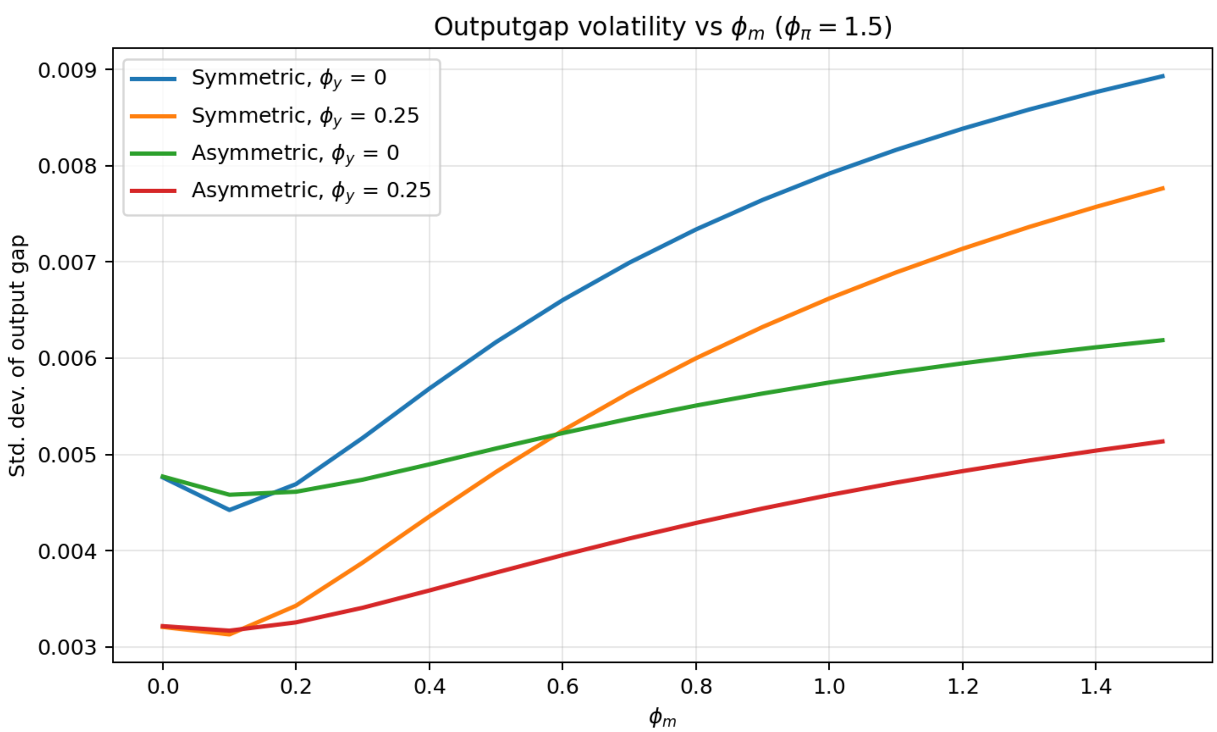


Figure 3: Output gap volatility different under monetary rules

Inflation outcomes are almost the exact opposite of asset price ones. In all settings the initial volatility response is mute, but quickly turns positive as ϕ_m increases. In light of our closed-form solutions, we can know that this is due to the increasing financial contamination of real outcomes that the asset price response creates. In line with this, the asymmetric rule induces less volatility in inflation, as it means the central bank is reacting to financial shocks less often. This is further supported by the impulse response functions in the next section.

Finally, output gap outcomes are the most dynamic. Under $\phi_y > 0$ the result is very similar to the ones found for inflation in 2. However, if there is no direct reaction to the output gap, a small positive ϕ_m produces *a lower* volatility of the output gap. Again the closed form results help inform why that would happen: insofar as asset prices move due to real shocks, reacting to them acts as a partial, imperfect substitute to reacting to output. When the central bank does not (or cannot) react to the real economy, weakly reacting to asset prices becomes worthwhile. This comes with the cost of introducing financial shocks into the economy, and the tradeoff quickly turns sour.

Taking stock we gather the following lessons from these figures:

First, since high ϕ_m delivers lower asset price volatility at the cost of higher output gap and inflation volatilities, it can be interpreted as a transfer of risk from financial assets to the real economy. Although our setup does not figure agent heterogeneity, in the real world this would be tantamount to insuring asset holding household returns at the expanse of non asset holding ones.

Second, a small positive ϕ_m is actually desirable. For small values, it has no impact on output gap and inflation outcomes, as the effects of increasing reactivity to real shocks cancel out those of introducing financial disturbance in the economy; but even at these small values it lowers asset price volatility, and with it the price of risk. A central bank concerned with overall smoothness in the economy would be keen to explore this opportunity.

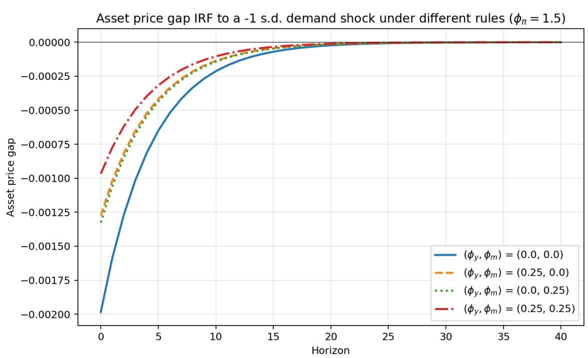
Third, that there are scenarios where positive, small ϕ_m not only delivers the above gains, but actually reduces output gap volatility. In the model, these are the ones when the central bank sets $\phi_y = 0$. In real life, we can think of situations when there is limited information on $\tilde{y}_t$ as ones where ϕ_y is forcibly set to zero. In these cases, asset prices may deliver valuable information, especially if the central bank can be reasonably sure the economy is undergoing a real shock. Recent examples of this could include the COVID-19 lockdown, which demanded monetary reactions months before any real activity data would be available; or the 2025 tariff and 2026 oil shocks, which have slow moving but undoubtedly negative effect on supply, but immediately readable effects on asset prices.

4.2 Impulse Response Functions

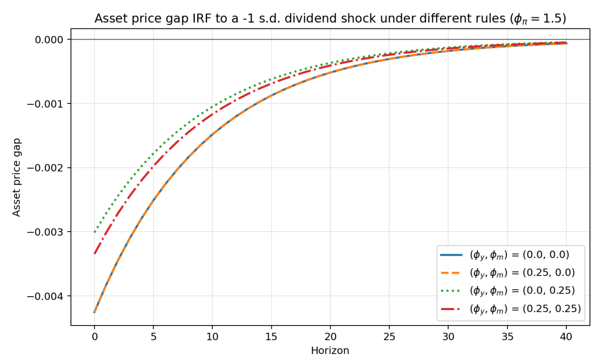
Impulse response functions add clarity to the previous results. Figures 4 through 6 make explicit the tradeoff of reacting to asset prices: *real* shocks in the form of demand and technology have reduced impacts on output gap, inflation and the asset price gap. *Financial* shocks in the form of dividends have reduced effects on asset prices, but become active for output and inflation, while they are neutral under $\phi_m = 0$

They also show how, for real shocks, reacting only to asset prices and inflation is near-equivalent to reacting to the output gap and inflation instead, highlighting the eventual usefulness in doing so.

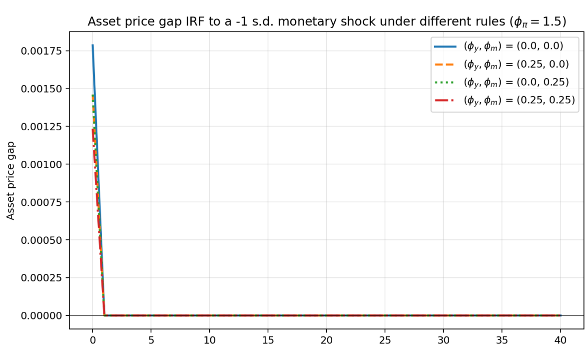
Taking together these help understand why the unconditional volatilities take the form they do as ϕ_m increases (that is, flat or U-shaped for small ϕ_m in inflation and output gap): The market-reacting rules dampen the inflation and output gap response of to real shocks, thus reducing their overall variability, and introduce a response to financial (here, dividend) shocks that were previously neutral, thus increasing it.



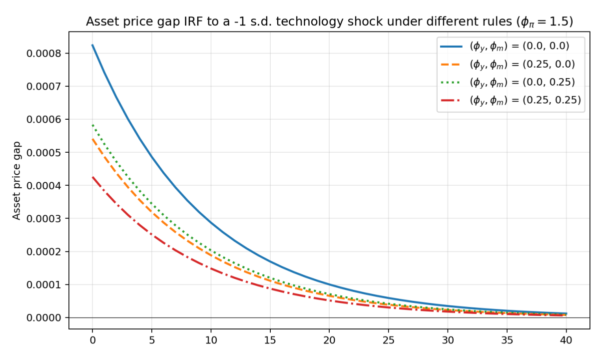
(a) Demand shock



(b) Dividend shock

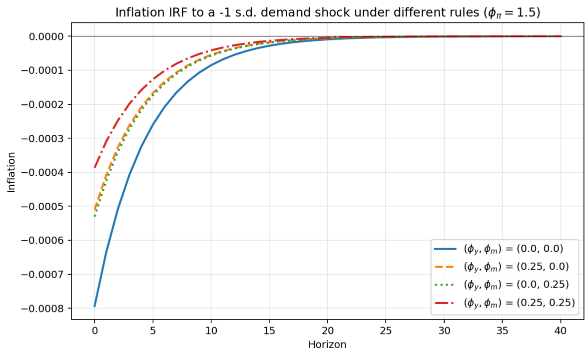


(c) Monetary shock

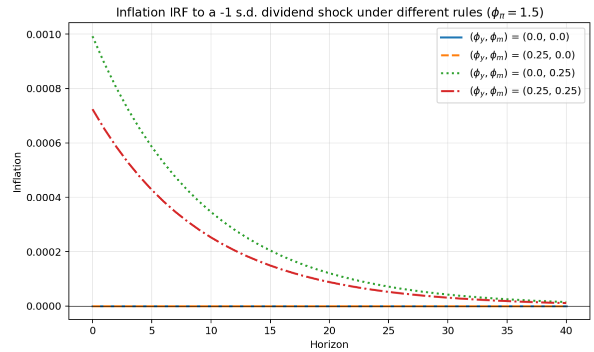


(d) Technology shock

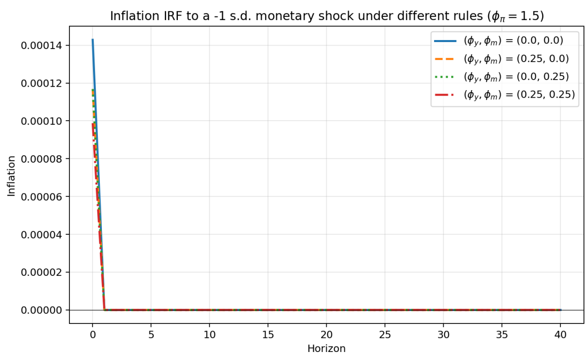
Figure 4: Impulse responses of asset price gap



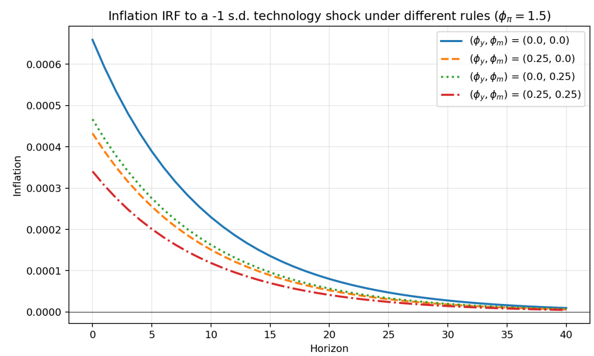
(a) Demand shock



(b) Dividend shock

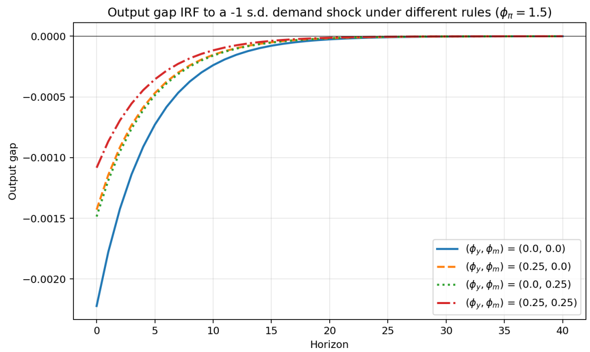


(c) Monetary shock

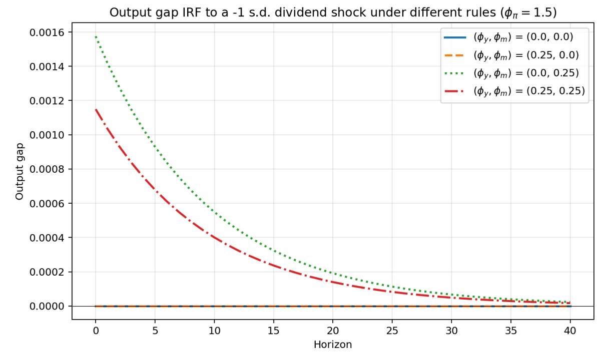


(d) Technology shock

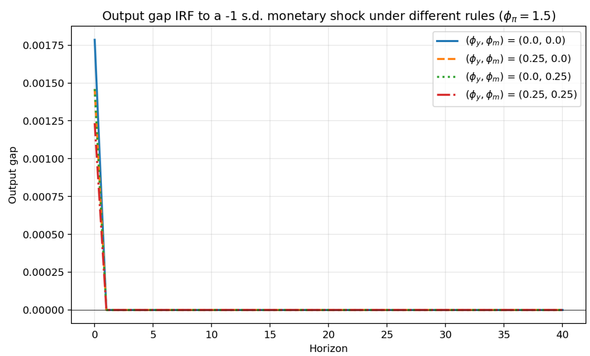
Figure 5: Impulse responses of inflation (π_t)



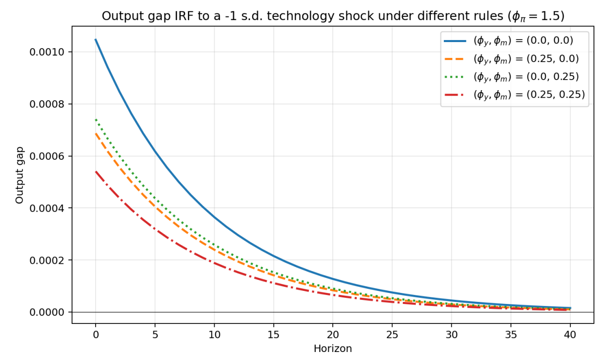
(a) Demand shock



(b) Dividend shock



(c) Monetary shock



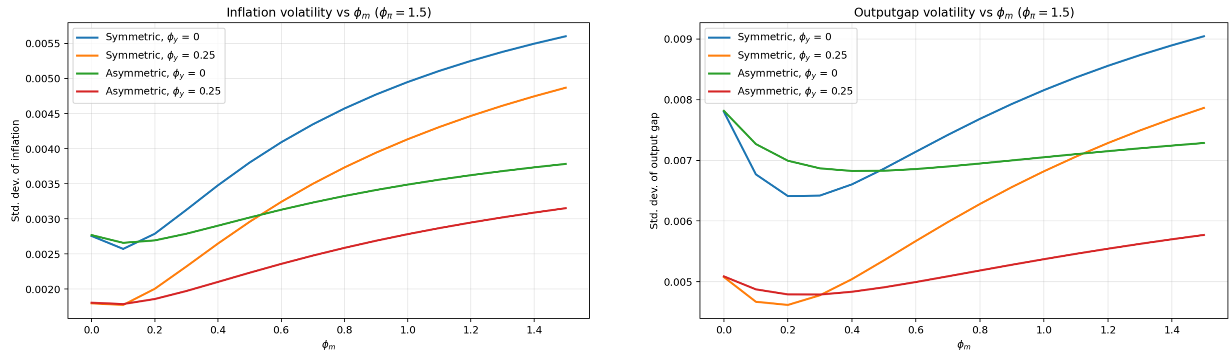
(d) Technology shock

Figure 6: Impulse responses of output gap

4.3 What of monetary shocks?

To allow for cleaner identification of the monetary rule effect on the economy, all of the previous exercises where conducted under $\rho_u = 0$, meaning monetary shocks have no persistence. Since the rule itself has no inertial term, this means monetary shocks are essentially trivialized.

By setting $\rho_u = 0.75$ we better approximate the persistence seen in interest rates. The results for economic volatility under different rules are qualitatively unchanged, but it has been said quantity has a quality of its own. As figure 7 makes clear, persistent monetary shocks significantly increase the value in reacting to asset prices in all settings. This highlights both the robustness and relevance of our previous results.



(a) Inflation

(b) Output gap

Figure 7: Economic volatility with persistent monetary shocks

5 Conclusion and Further Work

This draft set out to explore the effects of the Fed put in a simple new-keynesian production economy. Through a closed form solution for the symmetric rule we provide a better understanding of the mechanisms behind the seminal Bernanke & Gertler (2000) results. Then, via simulation methods we provide a more accurate treatment of the actual Fed put, i.e an asymmetric policy rule, and compare it to standard monetary rules.

These exercises cast doubt on those results. By exploring a more comprehensive set of parameters we see that there may be value in the put. At low sensitivities, responding to deviations in asset prices lowers asset price volatility while remaining neutral to inflation and output ones, thus delivering an overall smoother economy. Finally, we discuss scenarios in which adopting a small positive ϕ_m reduces financial and real economic volatility, namely when information on real output is for some reason unavailable.

We also show that the rule the literature usually explores is too harsh. A true "Fed put" would have the central bank reacting only to shortfalls in asset prices, not windfalls. By making the central bank more parsimonious, we show that this type of rule tends to generate better results than the symmetric one as measured by economic volatility.

In a full version of this paper we would seek to provide model determinacy results for the model, derive precise equivalencies between ϕ_m and ϕ_y for real shocks, and identify optimal rule parameters $(\phi_\pi, \phi_y, \phi_m)$. After that, two avenues suggest themselves:

The more immediate one is to do with how we are modeling the asset market. First, we could come closer to the production economy asset pricing literature by adding firm capital and pricing that instead of relying on the Lucas tree. This comes with the cost of losing control over financial shocks and parameters, and adding complications to a as-of-now very simple and treatable model. Second, permitting persistence in the asset price volatility would bring us closer to real market process. Third, as of now our setup cannot generate significant changes in mean asset prices due to different policy rules, even as this is one of the key aspects of the Fed put discussion. Overcoming this limitation would allow for interesting new results.

Another route is to explore agent heterogeneity. Adopting an OLG setup creates a channel through which the "tree" we planted in this economy directly affects consumption, and thus output. This would both enrich our results and allow for a wealth effect of market valuations to depress or enhance consumption. A mechanism like that would resemble the one the Fed believes acts in the American economy.

A Closed-form Solution of the Symmetric Model

As demand and technology shocks are summarized by the natural rate, we proceed by treating it as an exogenous process instead of treating each shock separately. This makes for a cleaner derivation throughout. Also in the spirit of simplification we solve the model under $d^{ss} = 0$.

Consider the linearized system

$$\tilde{y}_t = E_t \tilde{y}_{t+1} - \frac{1}{\sigma}\left(i_t - E_t \pi_{t+1} - r_t^n\right), \tag{27}$$

$$\pi_t = \beta E_t \pi_{t+1} + \kappa \tilde{y}_t, \tag{28}$$

$$\tilde{m}_t = \mu E_t \tilde{m}_{t+1} + (1-\mu) E_t \tilde{d}_{t+1} - \left(i_t - E_t \pi_{t+1} - r_t^n\right), \tag{29}$$

$$i_t = \phi_\pi \pi_t + \phi_y \tilde{y}_t + \phi_m \tilde{m}_t + u_t. \tag{30}$$

(We omit the steady-state constant from the policy rule, since it plays no role for deviations from steady state.)

Because the model is linear, one may solve it one shock at a time and then add the resulting components. Let s_t denote a generic AR(1) state,

$$s_t = \lambda s_{t-1} + \varepsilon_t, \qquad E_t s_{t+1} = \lambda s_t, \tag{31}$$

and suppose that it enters the model through

$$r_t^n = \eta s_t, \qquad u_t = \xi s_t, \qquad (1-\mu) E_t \tilde{d}_{t+1} = \nu \lambda s_t. \tag{32}$$

The three cases of interest are:

Shock	η	ξ	ν
Natural-rate shock r_t^n	1	0	0
Monetary policy shock u_t	0	1	0
Dividend shock d_t	0	0	$1-\mu$

where in the dividend case the state is $s_t = \tilde{d}_t$ itself.

Seek a solution of the form

$$\tilde{y}_t = A_y s_t, \qquad \pi_t = A_\pi s_t, \qquad \tilde{m}_t = A_m s_t. \tag{33}$$

Then

$$E_t \tilde{y}_{t+1} = \lambda A_y s_t, \qquad E_t \pi_{t+1} = \lambda A_\pi s_t, \qquad E_t \tilde{m}_{t+1} = \lambda A_m s_t. \tag{34}$$

Substituting the policy rule into the IS curve yields

$$A_y s_t = \lambda A_y s_t - \frac{1}{\sigma}\Big[(\phi_\pi - \lambda)A_\pi + \phi_y A_y + \phi_m A_m + \xi - \eta\Big] s_t, \tag{35}$$

so that

$$\big[\sigma(1-\lambda) + \phi_y\big] A_y + (\phi_\pi - \lambda)A_\pi + \phi_m A_m = \eta - \xi. \tag{36}$$

From the NKPC,

$$A_\pi = \beta\lambda A_\pi + \kappa A_y \quad \Longrightarrow \quad A_\pi = \frac{\kappa}{1-\beta\lambda} A_y. \tag{37}$$

Now define the real-rate gap

$$x_t \equiv i_t - E_t\pi_{t+1} - r_t^n. \tag{38}$$

Using the IS curve,

$$x_t = \sigma\left(E_t\tilde{y}_{t+1} - \tilde{y}_t\right) = -\sigma(1-\lambda)A_y s_t. \tag{39}$$

Substituting this into asset pricing gives

$$A_m s_t = \mu\lambda A_m s_t + \nu\lambda s_t + \sigma(1-\lambda)A_y s_t, \tag{40}$$

that is,

$$A_m = \frac{\nu\lambda + \sigma(1-\lambda)A_y}{1-\mu\lambda}. \tag{41}$$

Replacing (37) and (41) into (36) yields

$$\Big[\sigma(1-\lambda) + \phi_y\Big] A_y + (\phi_\pi - \lambda)\frac{\kappa}{1-\beta\lambda} A_y + \phi_m \frac{\nu\lambda + \sigma(1-\lambda)A_y}{1-\mu\lambda} = \eta - \xi. \tag{42}$$

Multiplying through by $(1-\beta\lambda)(1-\mu\lambda)$ and collecting terms in A_y gives

$$\begin{aligned} A_y\Bigg\{(1-\mu\lambda)\Big[\kappa(\phi_\pi-\lambda) + (1-\beta\lambda)\big(\phi_y + \sigma(1-\lambda)\big)\Big] + (1-\beta\lambda)\phi_m\sigma(1-\lambda)\Bigg\} \\ = (1-\beta\lambda)\Big[(1-\mu\lambda)(\eta-\xi) - \phi_m\nu\lambda\Big]. \end{aligned} \tag{43}$$

Define

$$\Delta(\lambda) \equiv (1-\mu\lambda)\Big[\kappa(\phi_\pi-\lambda) + (1-\beta\lambda)\big(\phi_y + \sigma(1-\lambda)\big)\Big] + (1-\beta\lambda)\phi_m\sigma(1-\lambda). \tag{44}$$

Then the output-gap coefficient is

$$A_y(\lambda;\eta,\xi,\nu) = \frac{(1-\beta\lambda)\Big[(1-\mu\lambda)(\eta-\xi) - \phi_m\nu\lambda\Big]}{\Delta(\lambda)}. \tag{45}$$

From (37),

$$A_\pi(\lambda;\eta,\xi,\nu) = \frac{\kappa\Big[(1-\mu\lambda)(\eta-\xi) - \phi_m\nu\lambda\Big]}{\Delta(\lambda)}. \tag{46}$$

Finally, substituting A_y back into (41), one obtains

$$A_m(\lambda;\eta,\xi,\nu) = \frac{\nu\lambda\Big[\kappa(\phi_\pi-\lambda) + (1-\beta\lambda)(\phi_y+\sigma(1-\lambda))\Big] + \sigma(1-\lambda)(1-\beta\lambda)(\eta-\xi)}{\Delta(\lambda)}. \tag{47}$$

Hence the coefficients under each shock are obtained by substitution:

$$\text{Natural-rate shock } (\lambda,\eta,\xi,\nu) = (\rho_r,1,0,0), \tag{48}$$
$$\text{Monetary shock } (\lambda,\eta,\xi,\nu) = (\rho_u,0,1,0), \tag{49}$$
$$\text{Dividend shock } (\lambda,\eta,\xi,\nu) = (\rho_d,0,0,1-\mu). \tag{50}$$

In particular,

$$A_y^{(r)} = \frac{(1-\beta\rho_r)(1-\mu\rho_r)}{\Delta(\rho_r)}, \qquad A_\pi^{(r)} = \frac{\kappa(1-\mu\rho_r)}{\Delta(\rho_r)}, \qquad A_m^{(r)} = \frac{\sigma(1-\rho_r)(1-\beta\rho_r)}{\Delta(\rho_r)}, \tag{51}$$

$$A_y^{(u)} = -\frac{(1-\beta\rho_u)(1-\mu\rho_u)}{\Delta(\rho_u)}, \qquad A_\pi^{(u)} = -\frac{\kappa(1-\mu\rho_u)}{\Delta(\rho_u)}, \qquad A_m^{(u)} = -\frac{\sigma(1-\rho_u)(1-\beta\rho_u)}{\Delta(\rho_u)}, \tag{52}$$

$$A_y^{(d)} = -\frac{(1-\beta\rho_d)\phi_m(1-\mu)\rho_d}{\Delta(\rho_d)}, \quad A_\pi^{(d)} = -\frac{\kappa\phi_m(1-\mu)\rho_d}{\Delta(\rho_d)}, \quad A_m^{(d)} = \frac{(1-\mu)\rho_d}{\Delta(\rho_d)}\Big[\kappa(\phi_\pi-\rho_d) + (1-\beta\rho_d)(\phi_y + \sigma(1-\rho_d))\Big] \tag{53}$$

B Parameters Used in Calibration

Here we provide our model parameters. At this stage we have calibrated directly the parameters that appear in the final log-linearized solution rather than the full primitive set of parameters.:

Table 1: Model Calibration

Parameter	**Description**	**Value**
Structural Parameters		
β	Discount factor	0.97
μ	Steady-state price-dividend ratio	0.97
σ	Risk aversion / intertemporal elasticity	1.00
κ	Phillips curve slope	0.08
ψ_{ya}^{n}	Natural rate semi-elasticity (technology)	1.00
Shock Persistences		
ρ_z	Demand shock persistence	0.80
ρ_a	Technology shock persistence	0.90
ρ_d	Dividend shock persistence	0.90
ρ_u	Monetary shock persistence	0.75
Shock Standard Deviations		
σ_z	Demand shock std. dev.	0.005
σ_a	Technology shock std. dev.	0.005
σ_d	Dividend shock std. dev.	0.020
σ_u	Monetary shock std. dev.	0.002